

Text Editor App in Python

This project is a Simple Text Editor built using Python's Tkinter GUI library, demonstrating file handling, menu creation, and event-driven programming.

□ Important Notes and Terms

□ Tkinter

Tkinter is Python's **built-in GUI (Graphical User Interface) library** used to create desktop applications with windows, buttons, menus, and text boxes.

□ Module

A **module** is a Python file that contains ready-made code.
Example:

□ `import tkinter as tk`

□ Here, `tkinter` is a module.

□ Submodule

A **submodule** is a smaller part inside a module that provides specific features.
Example:

□ `from tkinter import filedialog, messagebox`

- □ `filedialog` → file open/save windows
- `messagebox` → popup messages

□ `tk.Tk()`

Creates the **main application window**.
Without this, no GUI can exist.

□ `root.title()`

Sets the **title text** shown on the window's top bar.

❑ `root.geometry("800x600")`

Sets the **window size**

- 800 → width
- 600 → height

❑ Text Widget

❑ `tk.Text()`

❑ Creates a **multi-line text editor** where users can type, edit, and view text.

❑ `wrap=tk.WORD`

Ensures text wraps **by words**, not by characters.
Improves readability and looks professional.

❑ `pack(expand=True, fill=tk.BOTH)`

- `expand=True` → widget grows with window
- `fill=tk.BOTH` → fills width and height

❑ Function

A **function** is a reusable block of code that runs **only when called**.
Example:

❑ `def open_file():`

❑

❑ `text.delete(1.0, tk.END)`

Deletes **all text** from the text editor.

- `1.0` → beginning
- `tk.END` → end

☐ `text.insert(tk.END, data)`

Inserts text at the **end of the editor**.
Used to safely load file content.

☐ `with open(...)`

Used for **safe file handling**.
Automatically:

- Opens the file
- Closes the file after use

☐ File Modes

- `"r"` → read file
- `"w"` → write file (overwrites)

☐ File Dialog

☐ `filedialog.askopenfilename()`

☐ Opens system file browser to **select a file**.

☐ `filedialog.asksaveasfilename()`

☐ Opens system window to **save a file**.

☐ Message Box

☐ `messagebox.showinfo()`

☐ Displays **popup messages** for success, info, or alerts.

☐ Menu Bar

A **menu bar** is the top bar containing options like:

- File
- Edit
- Help

Created using:

□ `tk.Menu()`

□

□ Submenu

A **submenu** is a dropdown inside a menu.

Example:

□ File → New, Open, Save, Exit

□

□ `add_command()`

Links a **menu option** to a **function**.

□ `root.quit`

Closes the application safely.

□ Event-Driven Programming

The program waits for **user actions** (clicks, typing).

Functions run **only when events occur**.

□ `root.mainloop()`

Starts the **event loop**.

- Keeps the window open
- Listens for user actions
- Ends when window is closed

Full Code

```
□ # Import tkinter module for creating GUI applications
import tkinter as tk
```

```
# Import submodule filedialog for open/save dialogs and submodule
messagebox for popups
from tkinter import filedialog, messagebox
```

```

# ----- MAIN WINDOW -----

# Create main application window
root = tk.Tk()

# Set window title
root.title("Simple Text Editor")

# Set window size
root.geometry("800x600")

# ----- TEXT EDITOR AREA -----

# Create text editor area
text = tk.Text(
    root,                                # Parent window
    wrap=tk.WORD,                        # Wrap text by words (not characters)
    font=("Helvetica", 12)              # Font style and size
)

# Make text area fill the entire window
text.pack(expand=True, fill=tk.BOTH)

# ----- FILE FUNCTIONS -----

# Function to create a new file (clear all text)
def new_file():
    # Delete all text from the text box (from start to end)
    text.delete(1.0, tk.END)

# Function to open an existing text file
def open_file():
    # Open file dialog to select a text file
    file_path = filedialog.askopenfilename(
        defaultextension=".txt",          # Default file extension
        filetypes=[("Text Files", "*.txt")] # Allow only .txt files
    )

    # Check if a file is selected
    if file_path:
        # Open the selected file in read mode
        with open(file_path, "r") as file:

```

```

        # Clear old text from editor
        text.delete(1.0, tk.END)
        # Insert file content into the text editor
        text.insert(tk.END, file.read())

# Function to save the current text to a file
def save_file():
    # Open save file dialog
    file_path = filedialog.asksaveasfilename(
        defaultextension=".txt",          # Default file extension
        filetypes=[("Text Files", "*.txt")] # Allow only .txt files
    )

    # Check if file path is selected
    if file_path:
        # Open file in write mode
        with open(file_path, "w") as file:
            # Write text editor content into file
            file.write(text.get(1.0, tk.END))

        # Show success message after saving file
        messagebox.showinfo("Info", "File saved successfully!")

# ----- MENU BAR -----

# Create menu bar
menu = tk.Menu(root)

# Attach menu bar to the window
root.config(menu=menu)

# Create File menu
# Dropdown under File which has options like New, Open, Save, Exit
file_menu = tk.Menu(menu)

# Add File menu to menu bar
menu.add_cascade(label="File", menu=file_menu)

# Add New option to File menu
file_menu.add_command(label="New", command=new_file)

# Add Open option to File menu
file_menu.add_command(label="Open", command=open_file)

```

```
# Add Save option to File menu
file_menu.add_command(label="Save", command=save_file)

# Add a separator line in menu
file_menu.add_separator()

# Add Exit option to close the application
file_menu.add_command(label="Exit", command=root.quit)

# ----- RUN APPLICATION -----

# Run the application continuously
# Starts and keeps the window open
root.mainloop()
□
```